Machine Learning

Dataset Description – Student Performance ML Dataset

The dataset `student_performance_ml.csv` contains academic and behavioral information of students. The objective of this dataset is to predict whether a student will **Pass (1)** or **Fail (0)** based on various input features.

Each row in the dataset represents one student, and each column represents a measurable factor that may influence academic performance.

Features Description

- **StudyHours** – Number of hours a student studies per day.
- **Attendance** – Percentage of class attendance.
- **PreviousScore** – Marks obtained in the previous examination.
- **AssignmentsCompleted** – Number of assignments completed by the student.
- **SleepHours** – Average number of hours the student sleeps per day.
- **FinalResult** – Target variable (Output):
  - 1 → Pass
  - 0 → Fail

Objective of the Dataset

The goal is to:

- Analyze how different factors affect student performance.
- Build a Machine Learning model to predict whether a student will pass or fail.
- Understand concepts such as training, testing, accuracy, confusion matrix, overfitting, and model evaluation.

1. Import DecisionTreeClassifier from sklearn.
Create a model object and train it using `fit()`.

2. Use the trained model to predict results for X_test.
Display predicted values along with actual values.

3. Calculate model accuracy using `accuracy_score`.
Display the result in percentage format.

4. Generate confusion matrix using sklearn.
Display it using ConfusionMatrixDisplay.

Explain clearly:

- True Positive
- True Negative
- False Positive
- False Negative

5. Calculate:

- Training accuracy
- Testing accuracy

Compare both and comment whether the model is overfitting or underfitting.

6. Train three Decision Tree models with:

- max_depth = 1
- max_depth = 3
- max_depth = None

Compare their testing accuracies and write your observations.

7. Use the trained model to predict result for a student with:

- StudyHours = 6
- Attendance = 85
- PreviousScore = 66

Page 2 / 3

- AssignmentsCompleted = 7
- SleepHours = 7

Will the student Pass or Fail?

8. Write a single structured Python program that performs:

1. Dataset loading
2. Data analysis
3. Visualization
4. Train-test split
5. Model training
6. Prediction
7. Accuracy calculation
8. Confusion matrix generation
9. Final conclusion

Your code should include proper comments explaining each step.